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In [2]: # -----
# TASK 5: Exploratory Data Analysis (Titanic Dataset)
# -----

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Display settings
pd.set_option("display.max_columns", None)

# -----
# 1. Load Dataset
# -----

df = pd.read_csv("titanic.csv")
print("Dataset Loaded Successfully!")
df.head()

# -----
# 2. Basic Information
# -----

print("\n Dataset Info:")
print(df.info())

print("\n Statistical Summary:")
print(df.describe())

print("\n Missing Values:")
print(df.isnull().sum())

# -----
# 3. Univariate Analysis
# -----

plt.figure(figsize=(6,4))
sns.countplot(x='Survived', data=df)
plt.title("Survival Count")
plt.show()

plt.figure(figsize=(6,4))
sns.countplot(x='Sex', data=df)
plt.title("Passenger Gender Count")
plt.show()

plt.figure(figsize=(6,4))
sns.histplot(df['Age'], kde=True)
plt.title("Age Distribution")
plt.show()

# -----
# 4. Bivariate Analysis
# -----

plt.figure(figsize=(6,4))
sns.barplot(x='Sex', y='Survived', data=df)
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plt.title("Survival Rate by Gender")
plt.show()

plt.figure(figsize=(6,4))
sns.barplot(x='Pclass', y='Survived', data=df)
plt.title("Survival Rate by Passenger Class")
plt.show()

plt.figure(figsize=(6,4))
sns.scatterplot(x='Age', y='Fare', hue='Survived', data=df)
plt.title("Age vs Fare (Colored by Survival)")
plt.show()

# -----
# 5. Multivariate Analysis
# -----

plt.figure(figsize=(10,7))
numeric_df = df.select_dtypes(include=['number'])
sns.heatmap(numeric_df.corr(), annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()

sns.pairplot(df[['Survived', 'Age', 'Fare', 'Pclass']], hue='Survived')
plt.show()

# -----
# 6. Boxplots (Outlier Detection)
# -----

plt.figure(figsize=(6,4))
sns.boxplot(x=df["Fare"])
plt.title("Fare Boxplot")
plt.show()

plt.figure(figsize=(6,4))
sns.boxplot(x=df["Age"])
plt.title("Age Boxplot")
plt.show()

# -----
# 7. Summary of Insights
# -----

print("""
SUMMARY OF FINDINGS:

1. Females had a much higher survival rate than males.
2. Passengers in 1st class survived more than 2nd and 3rd.
3. Age distribution shows many passengers were young.
4. Fare has outliers—some passengers paid very high prices.
5. Heatmap shows strong correlation between Fare & Pclass.
6. Younger passengers and females had higher survival chances.
7. Missing values exist in Age and Cabin columns.
""")

```

Dataset Loaded Successfully!

Dataset Info:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 891 entries, 0 to 890

Data columns (total 12 columns):

#	Column	Non-Null Count	Dtype
0	PassengerId	891 non-null	int64
1	Survived	891 non-null	int64
2	Pclass	891 non-null	int64
3	Name	891 non-null	object
4	Sex	891 non-null	object
5	Age	714 non-null	float64
6	SibSp	891 non-null	int64
7	Parch	891 non-null	int64
8	Ticket	891 non-null	object
9	Fare	891 non-null	float64
10	Cabin	204 non-null	object
11	Embarked	889 non-null	object

dtypes: float64(2), int64(5), object(5)

memory usage: 83.7+ KB

None

Statistical Summary:

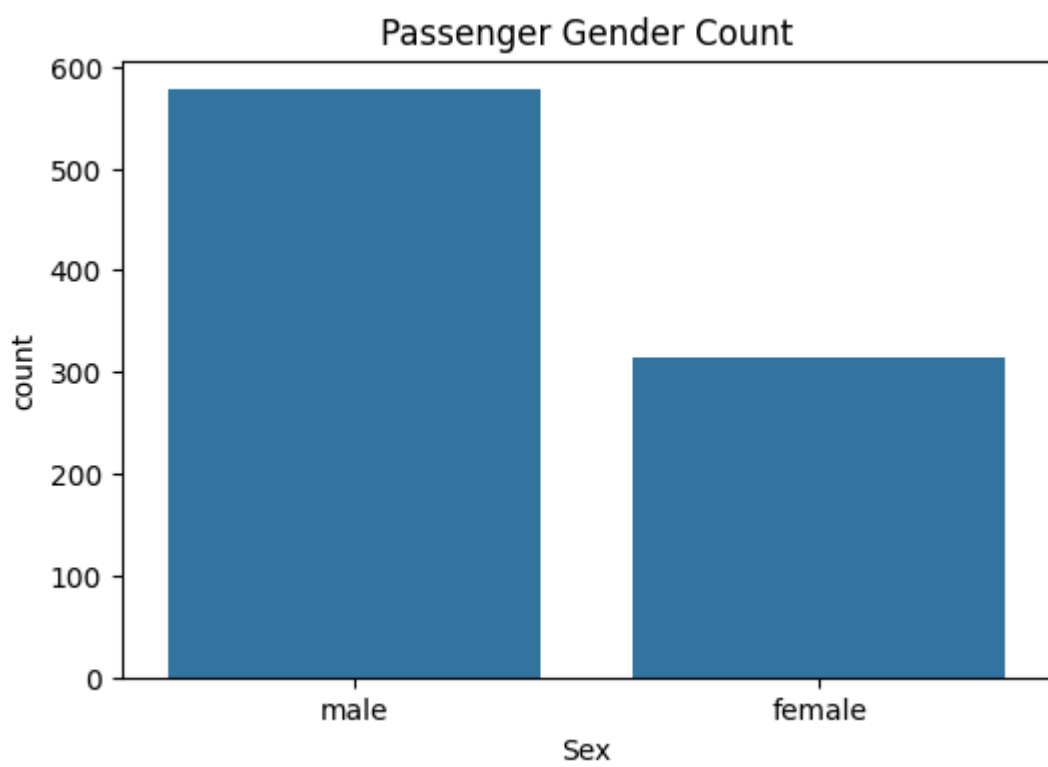
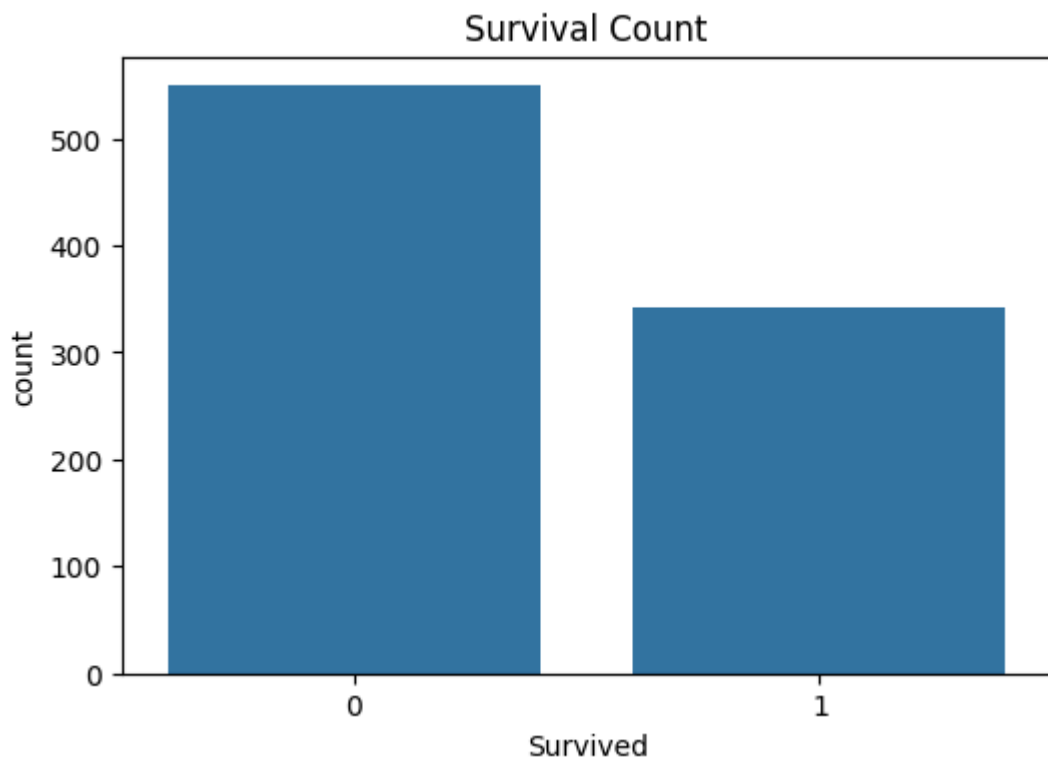
	PassengerId	Survived	Pclass	Age	SibSp \
count	891.000000	891.000000	891.000000	714.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008
std	257.353842	0.486592	0.836071	14.526497	1.102743
min	1.000000	0.000000	1.000000	0.420000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000

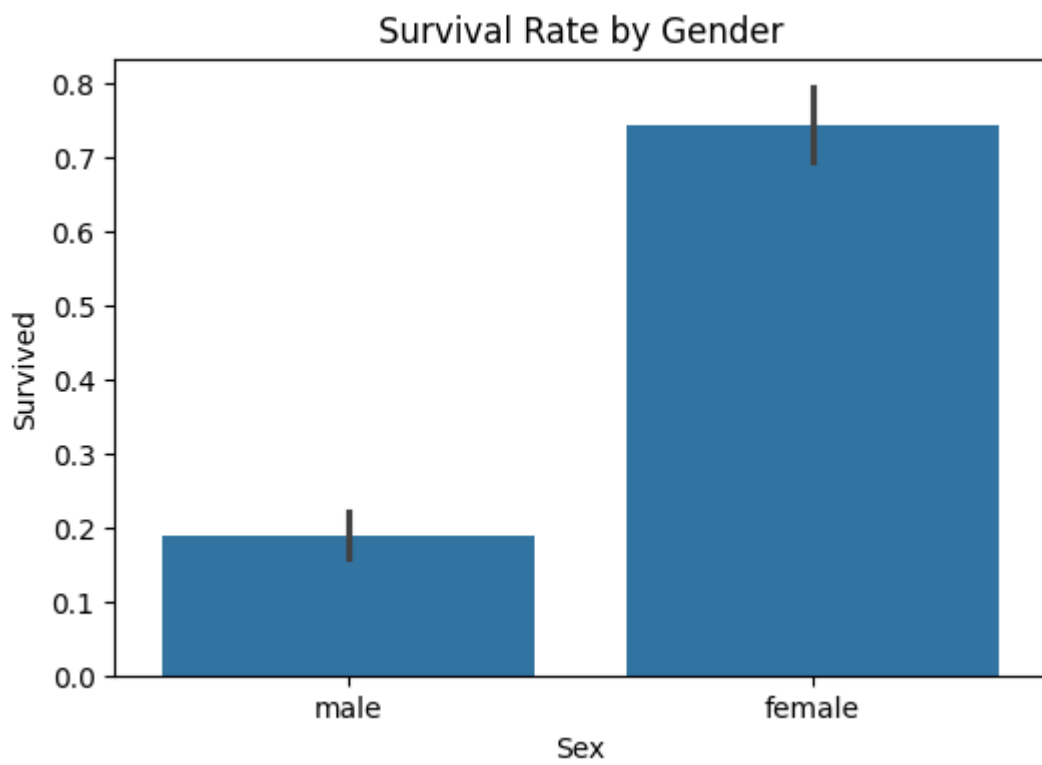
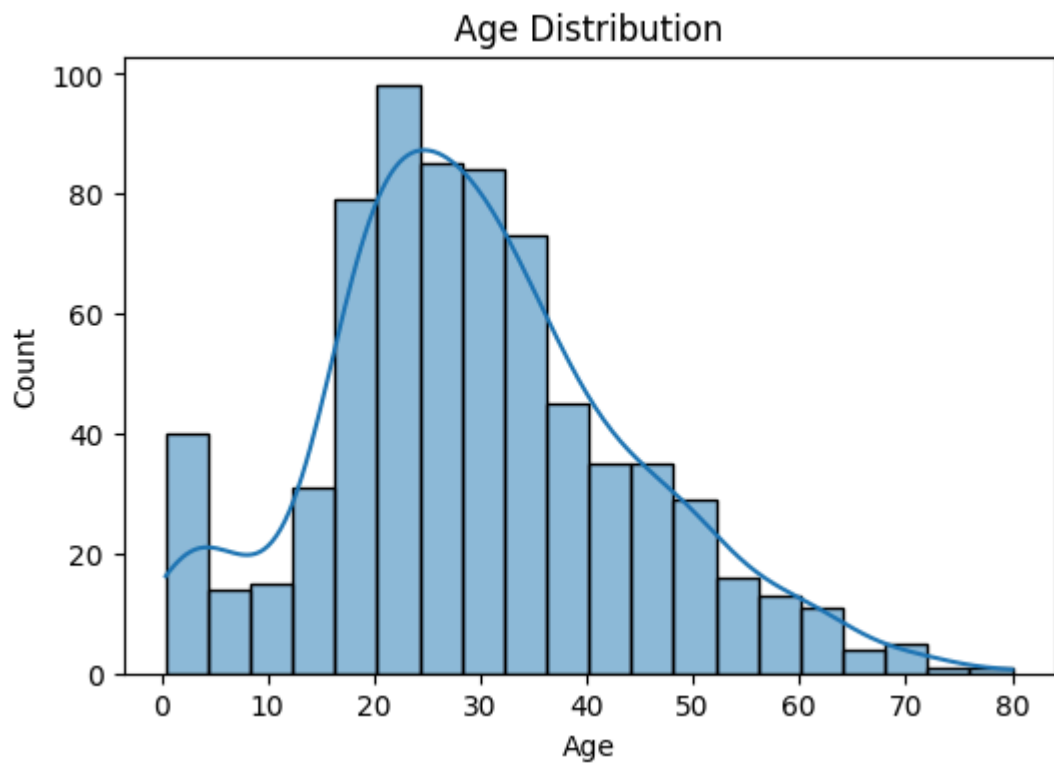
	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200

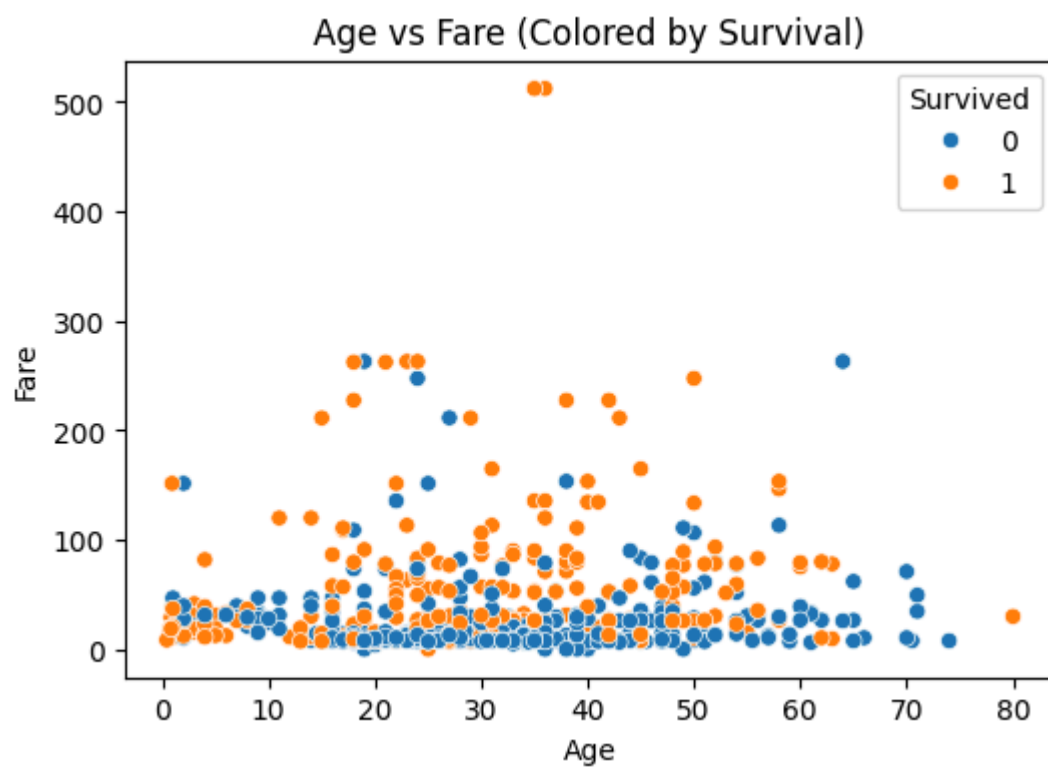
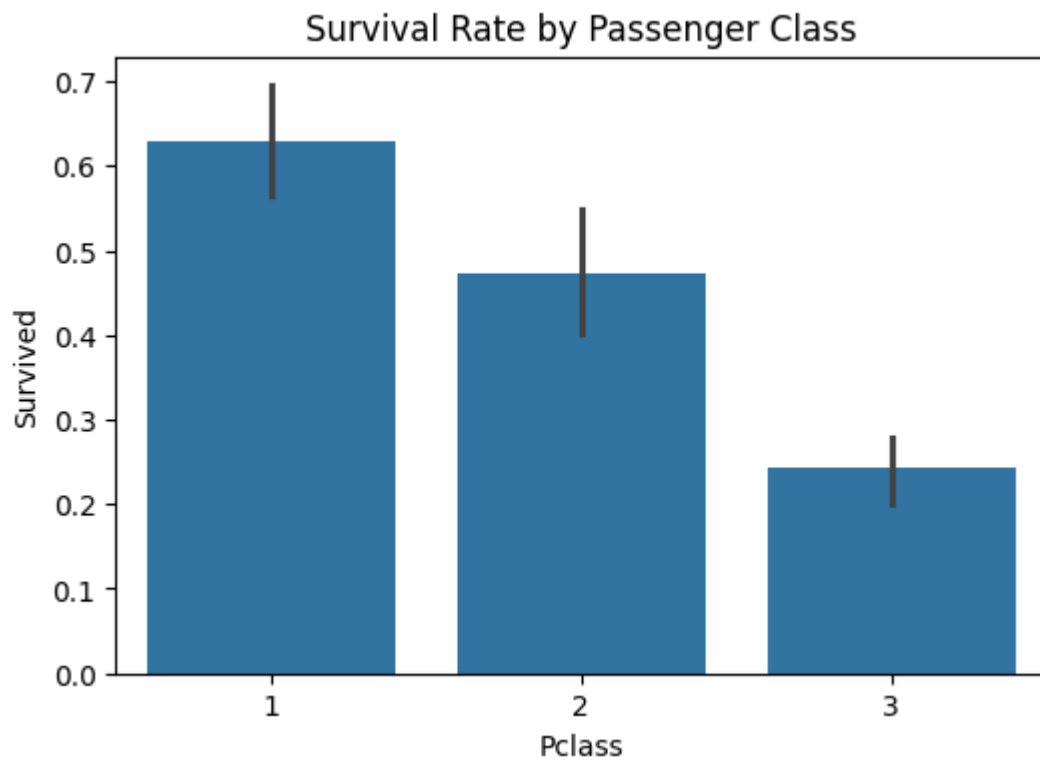
Missing Values:

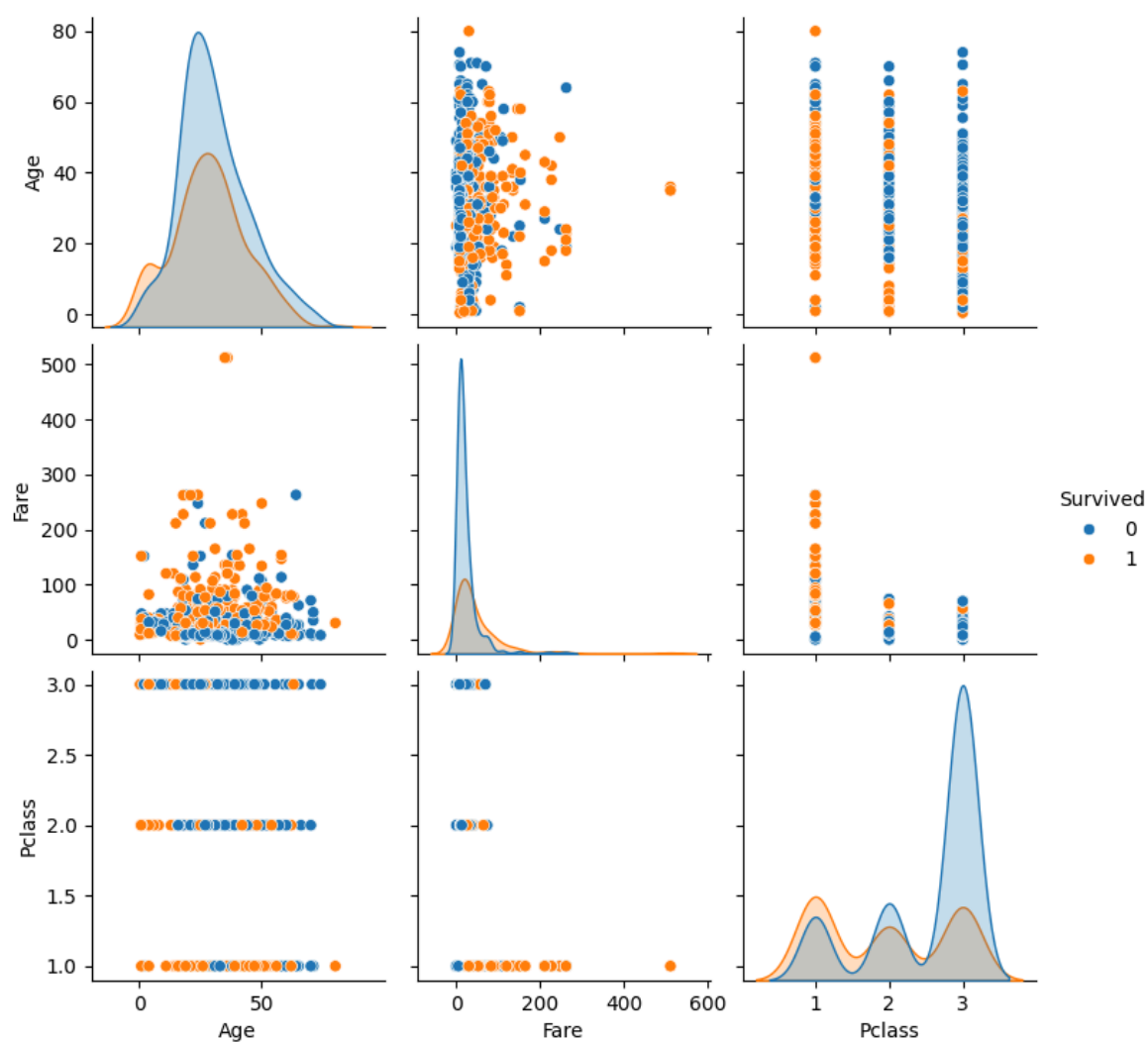
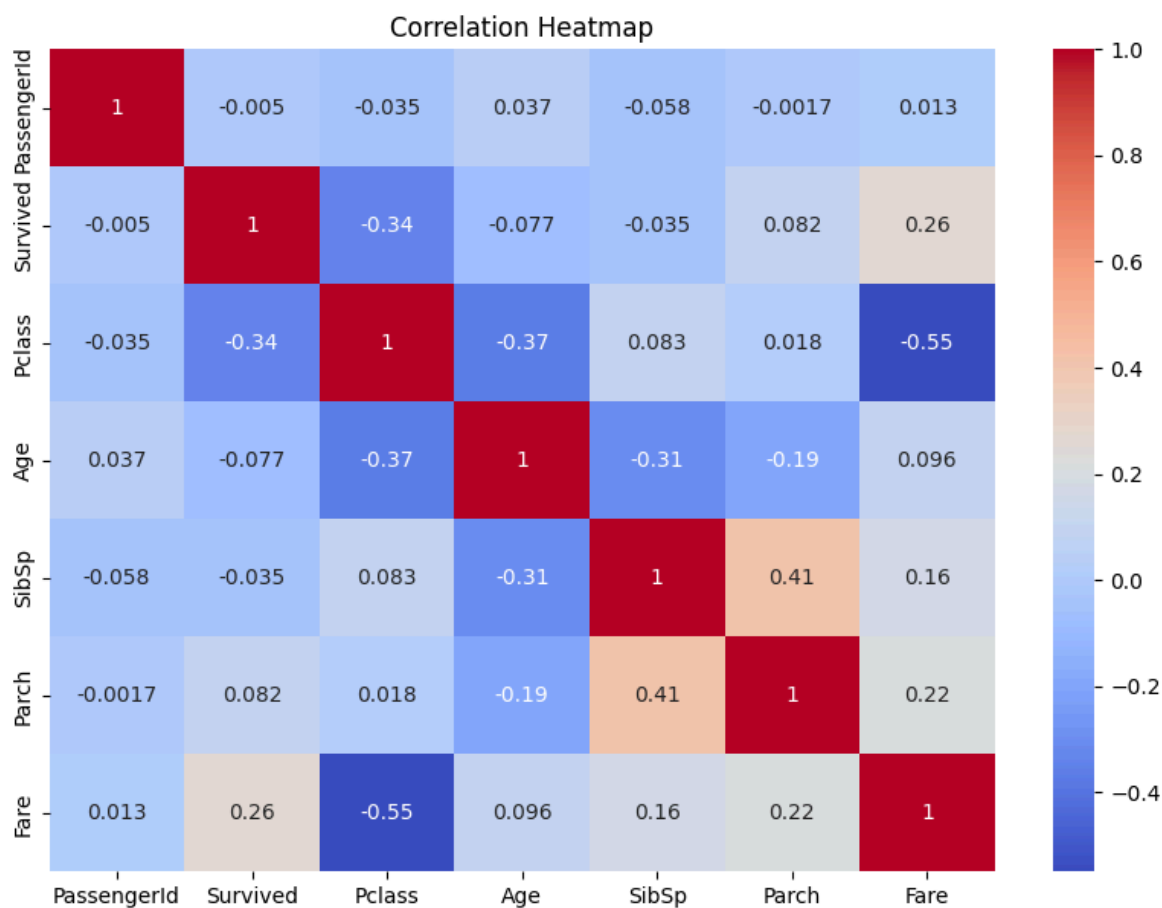
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

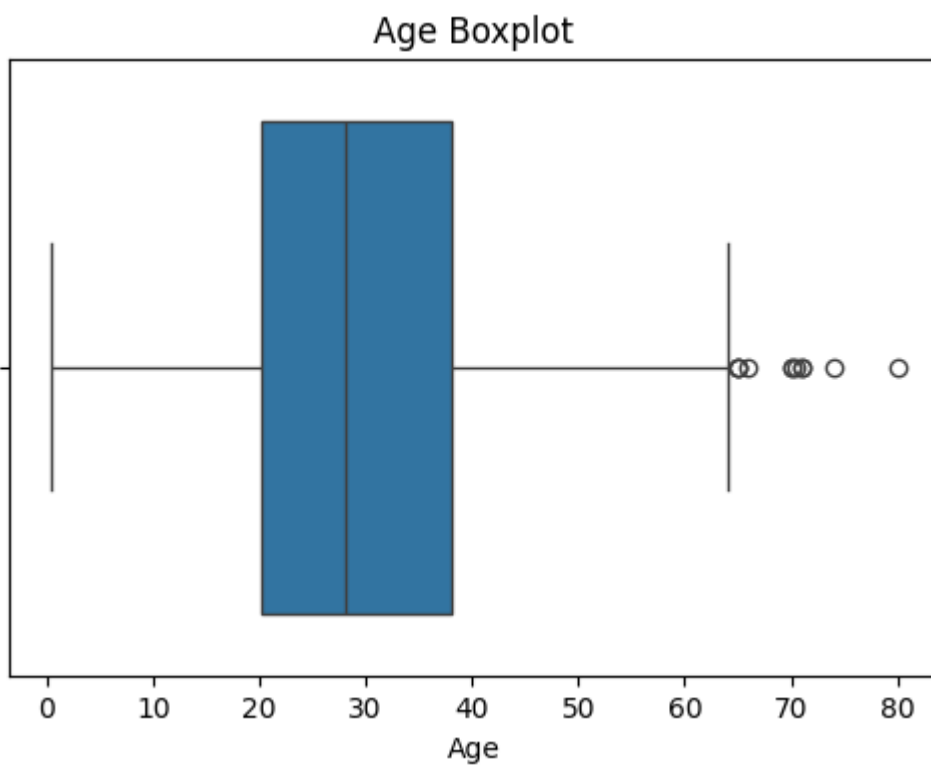
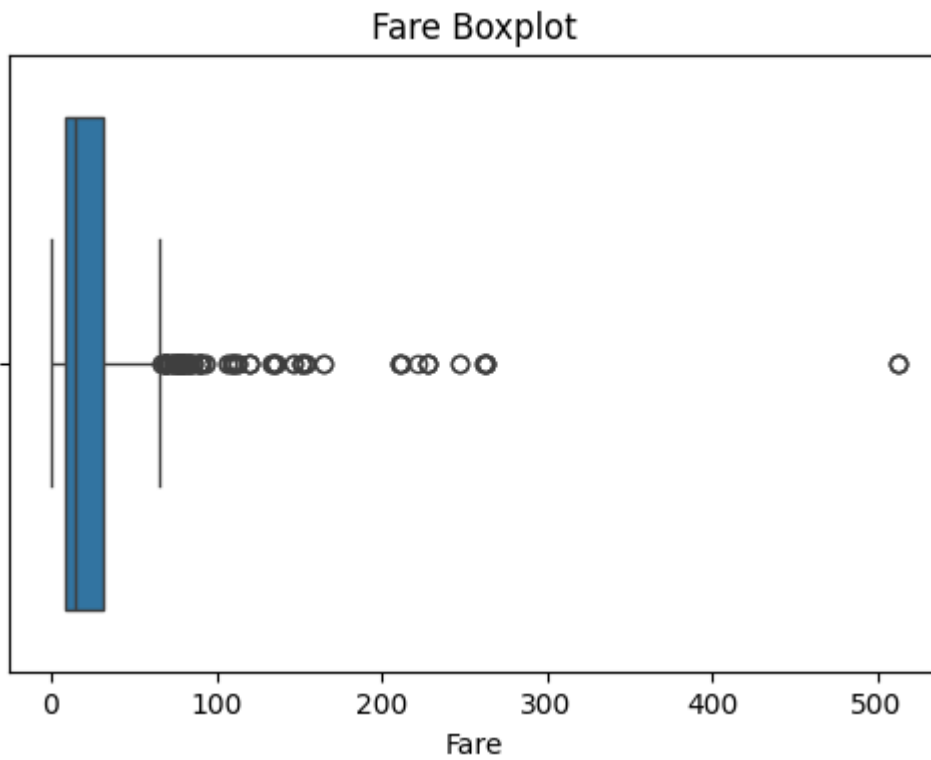
dtype: int64











SUMMARY OF FINDINGS:

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